# A Scalable Data Cleaning and Mining Framework for NYC Yellow Taxi Trip Data

**Affan Mohammed N M** **Gayathiri N** **Gopika Archa P J** **Rizwana Nazar R**

**2026EMAI10003** **2026EMAI10030** **2026EMAI10033** **2026EMAI10047**

**Department of Computer Science and Engineering**

**IIIT Kottayam, India**

## ABSTRACT

This paper presents a scalable data cleaning and mining framework applied to the New York City Yellow Taxi Trip dataset, comprising over 11.4 million records. A structured preprocessing pipeline is developed to address missing values, inconsistent data types, temporal anomalies, and extreme outliers. The pipeline improves data quality by removing 2.69% invalid records, resulting in a refined dataset suitable for analysis. Exploratory Data Analysis (EDA) reveals key temporal demand patterns and fare relationships. Furthermore, K-Means clustering is applied to identify three distinct trip behaviour segments: short urban commutes, medium-distance borough trips, and long-distance airport rides. The findings demonstrate that systematic preprocessing combined with unsupervised learning can effectively extract actionable insights from large-scale urban mobility data.

## I. INTRODUCTION

Urban transportation systems generate large-scale transactional data that can provide valuable insights into commuter behavior, demand patterns, and revenue dynamics. The New York City Taxi and Limousine Commission (TLC) publish monthly yellow taxi trip records, which include detailed attributes such as pickup and drop-off timestamps, geographic zone identifiers, fare components, payment types, and passenger counts. These datasets constitute a widely used and richly structured source of urban mobility data for research and policy analysis.

Despite their richness, raw taxi datasets often exhibit significant data quality issues. Missing values, inconsistent data types, temporal anomalies (e.g., trips where drop-off occurs before pickup), extreme outliers in distance and duration, and invalid passenger counts can adversely affect downstream analytical tasks. Without systematic preprocessing, analytical results may reflect data artifacts rather than genuine real-world patterns.

This study addresses these challenges by developing a structured data cleaning and mining framework applied to three months of NYC Yellow Taxi data. The primary objectives of this work are:

- Develop a scalable and systematic preprocessing pipeline for large-scale taxi trip data.
- Perform exploratory data analysis to uncover temporal and fare-related patterns.
- Apply clustering techniques to identify meaningful trip behavior segments.

The proposed framework ensures data quality while enabling robust analytical insights, contributing to improved understanding of urban mobility patterns and supporting data-driven decision-making in transportation systems.

## II. RELATED WORK

The NYC Taxi and Limousine Commission (TLC) dataset has been extensively studied in the data science and urban analytics community. Early research primarily focused on demand forecasting, employing time-series models to predict pickup volumes at the zone level [1]. Subsequent studies incorporated spatial analysis to identify demand hotspots and examine the influence of external factors such as weather conditions, public events, and holidays on taxi usage patterns [2].

From a data quality perspective, prior work has highlighted the impact of missing values and outliers on analytical outcomes. Zheng et al. [3] demonstrated that uncleaned data can significantly distort predictive models, particularly in fare estimation tasks. Common preprocessing strategies include interquartile range (IQR)-based outlier handling, statistical imputation techniques, and temporal consistency validation.

Clustering approaches have also been widely applied to taxi datasets to uncover latent trip patterns. Ferreira et al. [4] utilized clustering methods to identify spatial demand regions, while Yuan et al. [5] explored segmentation based on temporal and mobility features. These studies demonstrate the effectiveness of unsupervised learning in revealing meaningful travel behavior patterns.

However, many existing works do not explicitly document their preprocessing pipelines or quantify the impact of individual data cleaning steps, limiting reproducibility and interpretability. In contrast, this study presents a fully structured and transparent preprocessing framework, with quantified improvements at each stage, applied to a large-scale and recent dataset. This approach enhances both the reliability of the analysis and the reproducibility of the methodology.

## III. DATASET DESCRIPTION

### A. Dataset Overview

The dataset used in this study consists of NYC Taxi and Limousine Commission (TLC) Yellow Taxi Trip Records for December 2025, January 2026, and February 2026. The data was obtained from the official TLC open data portal in Apache Parquet format. The three-monthly datasets were merged into a single DataFrame containing 11,429,761 trip records and 20 feature columns.

### B. Feature Taxonomy

The dataset features are categorized into four types based on their characteristics and preprocessing requirements: temporal, categorical, numerical, and identifier features. This categorization guides the selection of appropriate data cleaning and transformation techniques.

| Type | Features | Notes |
|---|---|---|
| Temporal | tpep_pickup_datetime, tpep_dropoff_datetime | Stored as strings; converted to datetime64 |
| Categorical | VendorID, store_and_fwd_flag, RatecodeID, payment_type | Mixed types; require encoding/cleaning |
| Numerical | trip_distance, fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount, congestion_surcharge, Airport_fee, cbd_congestion_fee | Continuous; exhibit skewness & outliers |
| Identifiers | PULocationID, DOLocationID, passenger_count | Zone IDs; passenger_count has nulls & zeros |

*Table I: Feature Taxonomy of the NYC Yellow Taxi Dataset*

## IV. METHODOLOGY

### A. Data Cleaning Pipeline

The preprocessing pipeline follows a structured sequence of operations, with each step targeting a specific class of data quality issue. The pipeline was implemented in Python using the *pandas* library on Google Colab, processing all 11.4 million records in memory.

#### 1) Data Type Correction

The timestamp columns *tpep_pickup_datetime* and *tpep_dropoff_datetime* were initially stored as generic string objects in the merged dataset. These were converted to datetime64[ns] format using pd.to_datetime(), enabling temporal arithmetic operations required for trip duration computation.

The *store_and_fwd_flag* column exhibited mixed representations (‘N’, ‘Y’, and NaN). After removing extraneous whitespace and standardizing missing values, null entries were imputed with the modal value ‘N’.

Additionally, the *payment_type* column was cast to a categorical data type to optimize memory usage and improve computational efficiency.

#### 2) Missing Value Imputation

Several columns exhibited substantial missingness. Specifically, *passenger_count* and *RatecodeID* each contained 3,306,857 missing values (28.9%). These were imputed using the mode (most frequently occurring value) and subsequently cast to integer types.

The value *99* in *RatecodeID*, representing an undefined rate code, was treated as invalid and replaced with the modal value prior to imputation.

The *congestion_surcharge* and *Airport_fee* columns also contained 3,306,857 missing entries each. These were imputed with zero, based on the assumption that the absence of a surcharge corresponds to no additional fee.

#### 3) Negative Value Correction

Ten fare-related columns were examined for negative values, which are not meaningful in a billing context. These columns included: *fare_amount, extra, mta_tax, tip_amount, tolls_amount, improvement_surcharge, total_amount, congestion_surcharge, Airport_fee,* and *cbd_congestion_fee*.

All negative values in these fields were replaced with zero using element-wise clipping, ensuring consistency with domain constraints.

#### 4) Temporal Consistency Checks

A derived feature, *trip_duration*, was computed as the difference between drop-off and pickup timestamps, expressed in minutes:

$$\text{trip\_duration} = (\text{t\_dropoff} - \text{t\_pickup}) / 60$$

Trips where the drop-off timestamp preceded the pick-up (5 records, trip_duration < 0) were removed as logically inconsistent. Trips with exactly zero duration (143,684 records) were also removed, as they represent data entry errors rather than genuine journeys. Additionally, 118,797 trips with durations between 0 and 1 minute were removed, as sub-minute rides cannot correspond to valid metered taxi journeys.

### 5) Outlier Treatment:

Descriptive statistics revealed extreme outliers: the maximum *trip_distance* was 328,522 miles and the maximum *trip_duration* was 9,265 minutes, which are both physically impossible values. The 99.9th percentile capping strategy was adopted:

| Feature | Pre-cap Maximum | 99.9th Percentile Cap Applied |
|---|---|---|
| trip_distance | 328,522.20 miles | 30.14 miles |
| trip_duration | 9,265.15 minutes | 127.85 minutes |
| mta_tax | 4.00 (anomalous) | 0.50 (standard rate) |

*Table II: Outlier Capping Thresholds*

### 6) Passenger Count Validation:

44,850 trips recorded a *passenger_count* of 0, which is invalid for a revenue-generating taxi trip. These records were removed. After this step, the valid passenger count range was 1-9, consistent with TLC vehicle capacity regulations.

| Cleaning step | Records Removed | Remaining Records |
|---|---|---|
| Raw Dataset | — | 11,429,761 |
| Zero-duration trips removed | 143,684 | 11,286,077 |
| Negative-duration trips removed | 5 | 11,286,072 |
| Sub-minute trips removed | 118,797 | 11,167,275 |
| Zero-passenger trips removed | 44,850 | 11,122,425 |

*Table III: Data Cleaning Stage Impact*

Table III presents the impact of each data cleaning step. A significant portion of records was removed due to zero-duration and sub-minute trips, indicating inconsistencies in the raw dataset.

### B. Feature Engineering

Three temporal features were extracted from *tpep_pickup_datetime* to support demand analysis: **hour** (0-23), **day** (day name string, Monday-Sunday), and **month** (integer 1-12). The computed **trip_duration** feature was retained as a key numerical attribute for both EDA and clustering. These engineered features enable temporal decomposition of trip patterns without requiring external data sources.

### C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distributional characteristics, temporal patterns, and inter-feature relationships within the cleaned dataset. Visualisation techniques were employed to identify trends and validate the effectiveness of preprocessing.

**Distribution Analysis**

The distributions of key numerical features such as *trip_distance* and *trip_duration* were examined using histograms with kernel density estimation (KDE). Both features exhibit right-skewed distributions, indicating that the majority of trips are short-distance and short-duration, with a smaller number of long trips forming the tail.

**Temporal Analysis**

Temporal patterns were analysed using extracted features including hour, day, and month.
Pickup activity shows a clear bimodal pattern, with peaks during morning (8-9 AM) and evening (5-7 PM) commuting hours. Demand is lowest during early morning hours (3-5 AM).
Weekly patterns indicate higher trip volumes on Fridays and Saturdays, reflecting increased leisure activity, while Sundays exhibit comparatively lower demand.

**Correlation Analysis**

A Pearson correlation matrix was generated for numerical features to identify relationships between variables.
Strong positive correlations were observed between *fare_amount*, *total_amount*, *trip_distance*, and *trip_duration*, indicating that fare is primarily driven by trip length and duration.
Moderate correlation between *tip_amount* and *fare_amount* suggests proportional tipping behavior. Other features such as *congestion_surcharge* and *Airport_fee* showed weak correlation with trip distance, as they are location-based charges.

**K-Means Clustering**

Unsupervised clustering was applied to discover latent trip behavior segments. The feature set for clustering includes *passenger_count*, *trip_distance*, *fare_amount*, *extra*, *mta_tax*, *tip_amount*, *tolls_amount*, *improvement_surcharge*, *total_amount*, *congestion_surcharge*, *Airport_fee*, *cbd_congestion_fee*, *trip_duration*, *hour*, and *month*.

Prior to clustering, all features were standardised using z-score normalisation via *StandardScaler* to ensure that variables.with larger magnitudes did not dominate the distance metric. The optimal number of clusters was determined using the Elbow Method by evaluating the within-cluster sum of squares (inertia) for values of K ranging from 1 to 10. The elbow point was observed at K=3, indicating that three clusters provide a suitable balance between model complexity and variance explained.

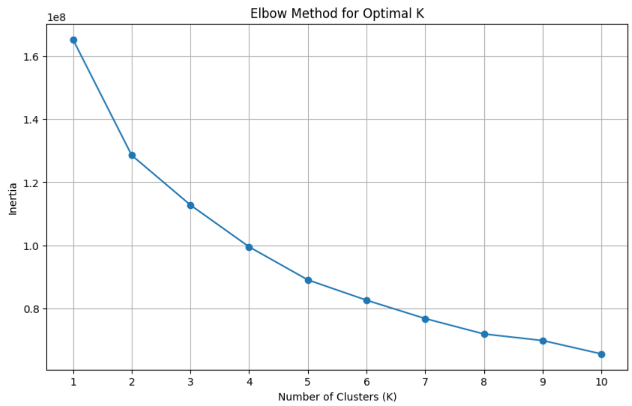



*Fig. 1: Elbow Method for Optimal Number of Clusters.*

Fig. 1 shows a clear inflection point at K=3, indicating the optimal number of clusters.

Based on this result, the K-Means algorithm was applied with K=3, using *random_state = 42* and *n_init = 10* to ensure reproducibility and stability of the clustering results.

| Cluster | No of Trips | Percentage |
|---|---|---|
| 0 | 6,375,534 | 57.3% |
| 1 | 3,796,003 | 34.1% |
| 2 | 844,482 | 7.6% |

*Table IV: Cluster Distribution*

## V. RESULTS AND ANALYSIS

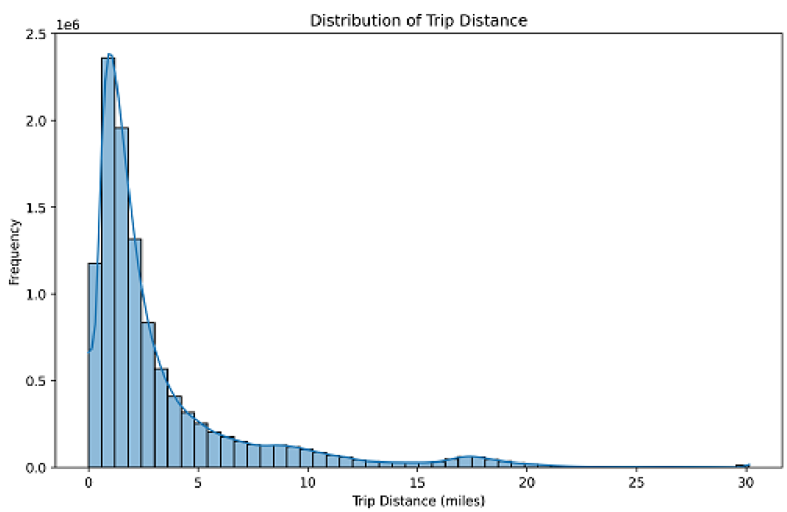
### A. Preprocessing Impact

The multi-stage cleaning pipeline systematically reduced the dataset from 11,429,761 to 11,122,425 records, resulting in a reduction of 307,336 records (2.69%). Table IV documents the row count after each cleaning stage.

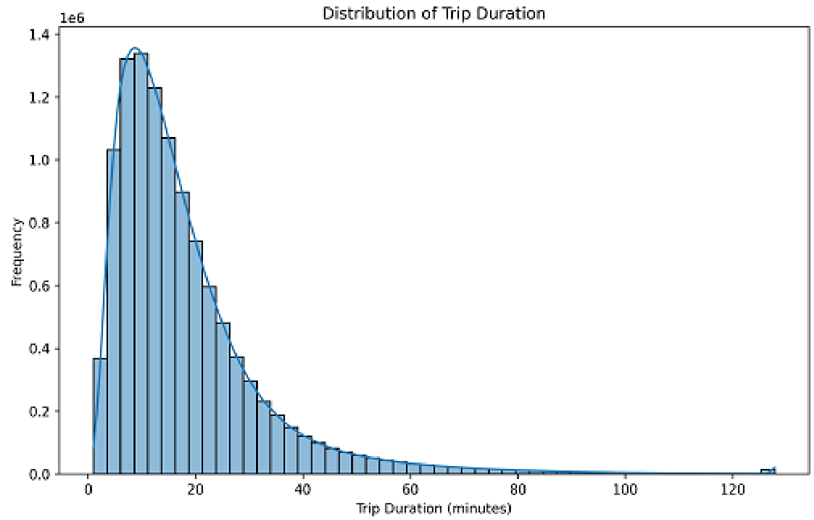
| Cleaning Stage | Records Removed | Remaining Records |
|---|---|---|
| Raw merged dataset | - | 11,429,761 |
| Zero-duration trips removed | 143,684 | 11,286,077 |
| Negative-duration trips removed | 5 | 11,286,072 |
| Sub-minute trips removed | 118,797 | 11,167,275 |
| Zero-passenger trips removed | 44,850 | 11,122,425 |
| **Final clean dataset** | **307,336 total (2.69%)** | **11,122,425** |

### B. Exploratory Data Analysis Findings

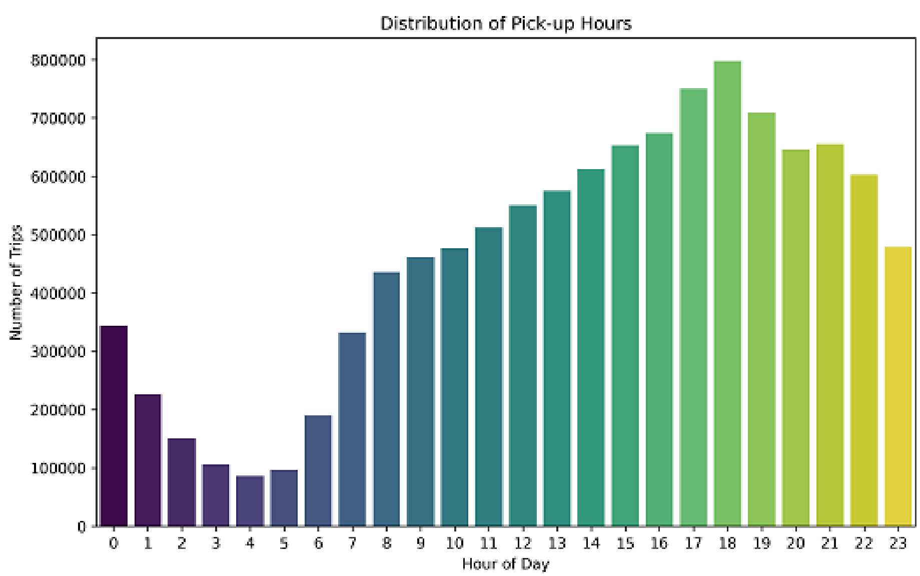
**Trip Distance Distribution**: Following capping at the 99.9th percentile (30.14 miles), trip distance shows a right-skewed distribution with the majority of trips falling within 0-5 miles, consistent with intra-borough urban commutes. The KDE overlay confirms a pronounced mode near 1 mile, reflecting short-haul demand typical of Manhattan.



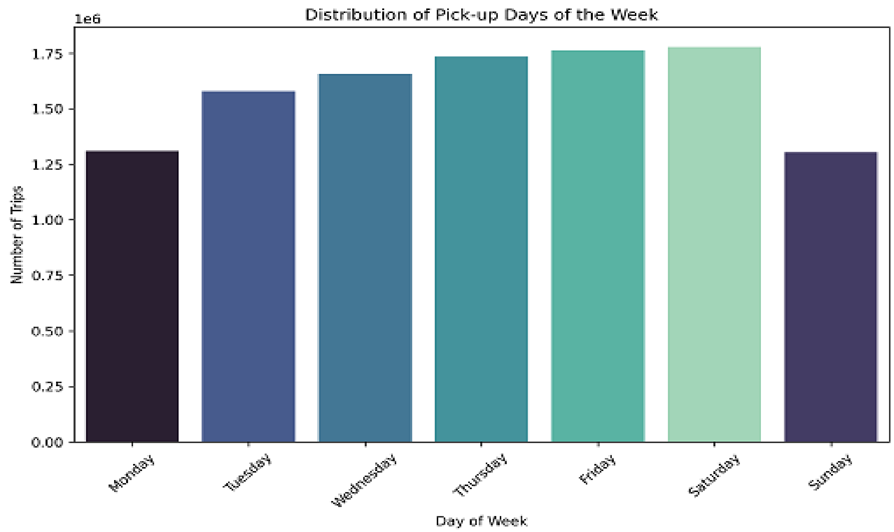
**Trip Duration Distribution:** Similarly skewed, trip duration peaks in the 5-15 minute range after capping at 127.85 minutes. The long tail represents cross-borough or airport trips. The bimodal structure visible in the raw distribution shows a secondary peak around 30-40 minutes and collapses post-capping, suggesting the secondary mode was driven by outliers.



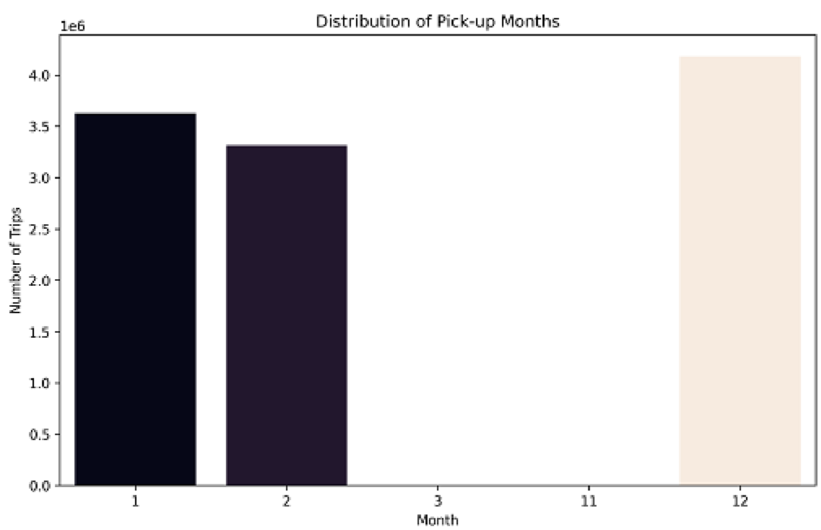

**Hourly Demand Patterns:** Pickup volume exhibits a bimodal daily pattern, with peaks at the morning rush (8-9 AM) and the evening rush (5-7 PM). Demand is lowest between 3-5 AM. This pattern is consistent across all three months, indicating stable commuter behaviour.



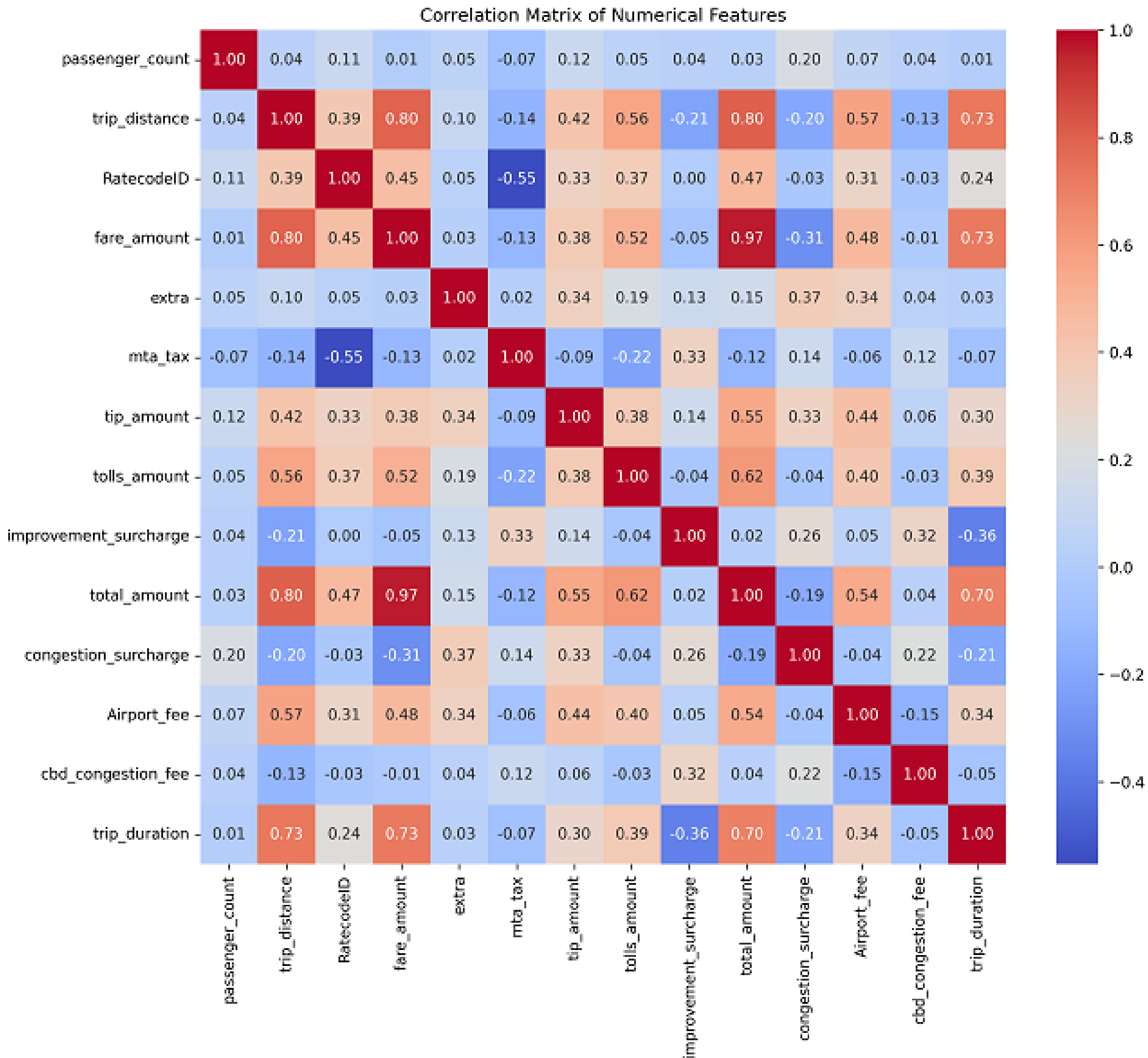
**Day-of-Week Patterns:** Friday and Saturday exhibit the highest trip volumes, driven by leisure and nightlife demand. Sunday shows the lowest weekday-equivalent volume. Weekday patterns (Monday-Thursday) are comparatively uniform.




**Monthly Patterns:** January 2026 recorded the highest trip volume among the three months, followed by December 2025 and February 2026. The February dip is partly attributable to the shorter calendar month.



**Correlation Analysis:** The Pearson correlation heatmap revealed strong positive correlations among fare components: *fare_amount*, *total_amount*, *trip_distance*, and *trip_duration* are highly intercorrelated (r > 0.75). *tip_amount* correlates moderately with *fare_amount* (r ≈ 0.58), reflecting tipping proportional to fare. *congestion_surcharge* and *Airport_fee* show low correlation with distance, indicating they are applied based on zone rather than trip length.



### C. Clustering Results

K-Means with K=3 converged on three well-separated clusters. Table V summarises the cluster sizes and their inferred behavioural profiles.

| Cluster | Size | % of Data | Profile | Key Characteristics |
|---|---|---|---|---|
| 0 | 6,375,534 | 57.3% | Standard Urban Commute | Short distance (1-3 mi), low fare, no airport fee, peak-hour pickups |
| 1 | 3,796,003 | 34.1% | Extended Borough Trip | Medium distance (3-10 mi), moderate fare, moderate duration, variable hours |
| 2 | 844,482 | 7.6% | Premium / Airport Ride | Long distance (10+ mi), high fare, Airport_fee applied, JFK/LGA routes |

*Table V: K-Means cluster profiles*

Cluster 0 is the dominant segment, capturing over half of all trips and characterising the routine intra-Manhattan short ride. Cluster 1 represents the intermediate category, likely outer-borough commutes or multi-stop journeys. Cluster 2, the smallest but economically significant segment, represents airport-bound rides with elevated fares and airport surcharges, constituting 7.6% of all trips but a disproportionately high share of total revenue due to fare magnitude.

Box plots of *trip_distance* across clusters confirmed monotonically increasing median distances from Cluster 0 to Cluster 2. Scatter plots of *trip_distance vs. trip_duration* showed near-linear within-cluster relationships, with Cluster 2 exhibiting higher variance in duration relative to distance, likely attributable to variable traffic conditions on routes to JFK and LaGuardia airports.

## VI. DISCUSSION

The preprocessing pipeline described in this study demonstrates that even a well-curated, officially published dataset requires substantial cleaning before analytical use. The 28.9% missing rate in *passenger_count* and *RatecodeID* were concentrated entirely in one vendor's records which suggests a systemic reporting issue rather than random missingness, warranting separate treatment from random MAR patterns.

Future work should investigate whether imputation with the global mode is appropriate, or whether vendor-specific imputation strategies would better preserve distributional fidelity.

The discovery of three behaviourally distinct trip clusters aligns with intuitive domain knowledge about NYC taxi usage. The practical implications are significant: ride-hailing platforms could use these clusters to design differentiated pricing tiers; fleet operators could allocate vehicles geographically based on cluster demand hotspots; and city planners could use demand peaks derived from Cluster 0 patterns to evaluate congestion pricing thresholds.

A limitation of the current approach is the use of K-Means, which assumes spherical, equally-sized clusters which is an assumption violated by the highly imbalanced cluster sizes observed (57% vs. 34% vs. 7.6%). Density-based methods such as DBSCAN or Gaussian Mixture Models may produce more nuanced segmentations, particularly for identifying anomalous or mixed-type trips at cluster boundaries.

## VII. LIMITATIONS AND CHALLENGES

Despite the effectiveness of the proposed framework, several practical challenges were encountered during the study.

First, the large volume of data (over 11.4 million records) posed computational constraints during preprocessing and analysis. Memory limitations in the working environment (Google Colab) occasionally resulted in kernel crashes, requiring careful optimization of operations and intermediate data handling.

Second, the dataset exhibited high dimensionality with multiple heterogeneous feature types. Determining which features to retain, transform, or exclude required iterative exploration and domain understanding, particularly to avoid introducing bias or redundancy into the analysis.

Third, handling missing values and outliers presented a significant challenge. While mode imputation and percentile-based capping were adopted, these approaches may not fully preserve the underlying data distribution, especially in cases of non-random missingness.

Additionally, the selection of appropriate visualization techniques required careful consideration. Inappropriate graph types or poorly chosen axis scales can misrepresent patterns and potentially mislead stakeholders, emphasizing the importance of effective visual communication in data analysis.

These challenges highlight the trade-offs between scalability, computational feasibility, and analytical precision when working with large-scale real-world datasets.

## VIII. CONCLUSION

This paper presented a structured data cleaning and mining framework for the NYC Yellow Taxi Trip dataset covering December 2025 through February 2026. A six-stage preprocessing pipeline resolved data type inconsistencies, imputed missing values, corrected negative fares, removed temporally invalid records, capped extreme outliers, and eliminated zero passenger entries, resulting in a high-quality dataset of 11,122,425 trips.

Exploratory analysis confirmed well known patterns such as morning and evening rush peaks and higher demand on Fridays and Saturdays, and also revealed strong relationships between fare and trip distance. K-Means clustering with K equal to 3 identified three interpretable trip archetypes corresponding to short urban commutes, extended borough trips, and premium airport rides.

These findings demonstrate that systematic data preprocessing is not merely a preliminary step but a substantive research contribution in its own right, particularly at scale. Future work will extend this framework to incorporate spatial clustering using geographic zone identifiers, apply time series forecasting for hourly demand prediction, and evaluate deep learning architectures for fare estimation on the cleaned dataset.

## IX. FUTURE WORK

Several directions remain open for investigation: -

First, spatial analysis incorporating *PULocationID* and *DOLocationID* zone geometries could enable geographic demand mapping and hotspot detection at sub-borough granularity.

Second, time-series models (ARIMA, Prophet, LSTM) applied to the hourly trip counts could support real-time demand forecasting for fleet dispatch optimisation.

Third, fare prediction using regression models trained on the cleaned dataset could benchmark the predictive improvement attributable to the preprocessing pipeline.

Finally, longitudinal analysis extending the dataset across multiple years would enable seasonality decomposition and post-pandemic recovery trend identification.

## REFERENCES

[1] Y. Tong, L. Chen, Z. Zhou, H. V. Jagadish, L. Shou, and W. Lv, "SIMBA: Scalable inversion-based moving objects detection in bigdata," in *Proc. VLDB*, 2014.

[2] N. Ferreira, J. Poco, H. T. Vo, J. Freire, and C. T. Silva, "Visual exploration of big Spatio-temporal urban data: A study of New York City taxi trips," *IEEE Trans. Vis. Comput. Graph.*, vol. 19, no. 12, pp. 2149-2158, 2013.

[3] C. Zheng, X. Fan, C. Wang, and J. Qi, "GMAN: A graph multi-attention network for traffic prediction," in *Proc. AAAI*, 2020.

[4] N. Ferreira et al., "Urbane: A 3D framework to support data driven decision making in urban development," in *Proc. IEEE VAST*, 2015.

[5] J. Yuan, Y. Zheng, and X. Xie, "Discovering regions of different functions in a city using human mobility and POIs," in *Proc. ACM KDD*, 2012.

[6] scikit-learn developers, "scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825-2830, 2011.

[7] NYC Taxi and Limousine Commission, "TLC Trip Record Data," [Online]. Available:
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

[8] W. McKinney, "Data structures for statistical computing in Python," in *Proc. SciPy*, 2010.